A MAP OF MODULES OVER A LOCAL RING WHOSE REDUCTION IS INJECTIVE IS UNIVERSALLY INJECTIVE

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Allowed standard prerequisites (editorial).

- A projective module over a local ring is free, and Lazard's theorem on flat modules as filtered colimits of finite free modules. Algebra, labels algebra-theorem-projective-free-over-local-ring (proof line 1955) and algebra-theorem-lazard (proof line 1960). The first reduces to the case of a finite free source, and the second writes the target as a directed colimit of finite free modules.
- Splitting of an injection of finite free modules over a local ring detected modulo the maximal ideal. Algebra, elementary fact used at proof lines 1967-1968. Shows that the reduction of the map being injective implies that the map into each sufficiently late finite free piece is a split injection, which is what makes it stay injective after tensoring with any module.

Source reference. Original theorem 058G: sources/stacks-chow/tags.txt, source lines 6779–6779. Navigation pointer for \ref{algebra-theorem-lazard} used inside the counted span (line 1960); the argument that consumes it is retained in the span.

Source reference. Original theorem 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. Navigation pointer for \ref{algebra-theorem-projective-free-over-local-ring} used inside the counted span (line 1955); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a local ring. Let $u : M \to N$ be an R-module map. If M is a projective R-module, N is a flat R-module, and $\overline{u} : M/\mathfrak{m}M \to N/\mathfrak{m}N$ is injective then u is universally injective.*

Proof. By Algebra, Theorem 0593 the module M is free. If we show the result holds for every finitely generated direct summand of M, then the lemma follows.

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Hence we may assume that M is finite free. Write $N = \operatorname{colim}_i N_i$ as a directed colimit of finite free modules, see Algebra, Theorem 058G. Note that $u : M \to N$ factors through N_i for some i (as M is finite free). Denote $u_i : M \to N_i$ the corresponding R-module map. As $\overline{u}$ is injective we see that $\overline{u_i} : M/\mathfrak{m}M \to N_i/\mathfrak{m}N_i$ is injective and remains injective on composing with the maps $N_i/\mathfrak{m}N_i \to N_{i'}/\mathfrak{m}N_{i'}$ for all $i' \geq i$. As M and $N_{i'}$ are finite free over the local ring R this implies that $M \to N_{i'}$ is a split injection for all $i' \geq i$. Hence for any R-module Q we see that $M \otimes_R Q \to N_{i'} \otimes_R Q$ is injective for all $i' \geq i$. As $- \otimes_R Q$ commutes with colimits we conclude that $M \otimes_R Q \to N_{i'} \otimes_R Q$ is injective as desired. $\square$